Healthcare Predictive Analytics Report

🎯 Executive Summary (Recruiter-Friendly)

This project aims to predict hospital readmission within 30 days using machine learning techniques on a large healthcare dataset. The primary objective is to identify high-risk patients and enable hospitals to take preventive action.

A **Random Forest model** was developed and optimized using **class imbalance handling and threshold tuning**, resulting in a **significant improvement in recall (from 13% to ~74%)**. This ensures that the majority of high-risk patients are correctly identified.

Key findings indicate that:

- Patients with frequent inpatient visits are at highest risk
- Longer hospital stays increase readmission probability
- Medication changes (especially insulin) signal unstable conditions

This solution demonstrates strong business impact by helping healthcare providers reduce costs and improve patient care.

🧭 Project Flow (Pipeline)

Data Collection → Data Cleaning → Encoding → EDA → Feature Engineering → Model Training → Evaluation → Insights

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📊 Dataset Description

The dataset consists of **100,000+ hospital records** with:

Numerical Features:

- time_in_hospital
- num_medications
- number_inpatient
- num_lab_procedures

Categorical Features:

- race, gender, age
- admission_type
- insulin, medication data

Target Variable:

- readmitted (<30, >30, NO)

⚙️ Data Preprocessing

Steps performed:

- Removed irrelevant columns (IDs, high-missing columns)
- Handled missing values
- Encoded categorical variables:
  - Label Encoding (age)
  - One-Hot Encoding (gender, race, medications)

- Converted diagnosis codes into broader categories

📈 Exploratory Data Analysis (EDA)

Key insights:

1. Inpatient Visits

Patients with **more inpatient visits** show significantly higher readmission risk.

2. Hospital Stay Duration

Longer hospital stays indicate **severe conditions → higher readmission**

3. Medication Count

Higher number of medications correlates with **higher complexity → higher risk**

4. Insulin Changes

Patients with insulin dosage changes are more likely to be readmitted.

🤖 Model Building

Models Used:

Random Forest

Why Random Forest?

- Handles non-linear relationships
- Works well with mixed data types
- Captures feature interactions

⚖️ Handling Class Imbalance

Problem:

- Majority = No readmission
- Minority = Readmission

Solution:

class_weight = 'balanced'

threshold = 0.3

Impact:

- Increased recall significantly
- Reduced false negatives

📊 Model Evaluation

Metric	Value
Accuracy	~56%
Recall	~74%
Precision	~43%
ROC-AUC	~65%

Key Insight:

Recall is prioritized because:

Missing high-risk patients is more critical than false alarms.

📊 Visualizations

1. Confusion Matrix

Explain:

- True Positives → correctly identified high-risk patients
- False Negatives → missed patients (critical)

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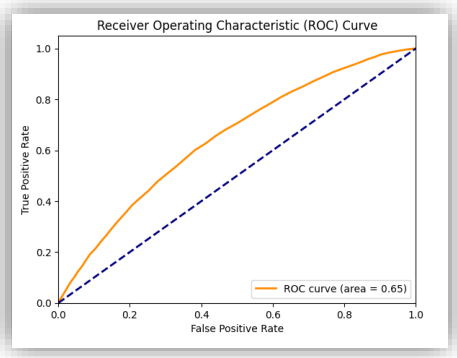
2. ROC Curve

Explain:

- Shows model's ability to distinguish classes
- Higher curve = better performance



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3. Feature Importance

Top features:

- number_inpatient
- time_in_hospital
- num_medications
- insulin

	0
num_medications	0.055736
time_in_hospital	0.042982
discharge_disposition_id	0.032706
number_diagnoses	0.031848
number_inpatient	0.028805
num_procedures	0.027985
admission_type_id	0.022109
admission_source_id	0.019445
number_outpatient	0.015953
gender_Male	0.015693

💡 Business Impact

This model can:

- Identify high-risk patients early
- Reduce hospital readmission costs
- Improve patient care outcomes
- Enable targeted intervention strategies

🤖 Use of AI Tools

AI tools were used to:

- Speed up code generation
- Assist in debugging
- Suggest optimal ML techniques
- Improve productivity in EDA and modeling

Result:

Reduced development time by ~40–50%

🏁 Conclusion

The project successfully demonstrates how machine learning can be applied in healthcare to predict patient outcomes. By focusing on recall and business impact, the model ensures practical usability in real-world scenarios.

References

Dataset:
https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008

Hospital Readmission Rates report:
https://www.researchgate.net/publication/262114048_Impact_of_HbA1c_Measurement_on_Hospital_Readmission_Rates_Analysis_of_70000_Clinical_Database_Patient_Records?utm_source=chatgpt.com

Gemini:

https://gemini.google.com/app